

YOUNGHYO PARK

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[Google Scholar](#)

EDUCATION

Massachusetts Institute of Technology

Sep. 2023 – Present

Ph.D. in Electrical Engineering and Computer Science; Advisor: Professor Pulkit Agrawal

Cambridge, MA

Seoul National University

Mar. 2016 – Aug. 2022

B.S. in Mechanical Engineering; GPA 4.25/4.3; summa cum laude; 1st in department

Seoul, South Korea

Completed mandatory military service, 2018–2019.

PUBLICATIONS

* denotes equal contribution.

- **Younghyo Park**, Antonia Bronars, and Pulkit Agrawal. *When Robots Outweigh Objects: How Inertia Mismatch Affects Robot Policy Learning*. Under review for the IEEE International Conference on Robotics and Automation (ICRA), 2027.
- Antonia Bronars*, **Younghyo Park***, Kushal Kedia, and Pulkit Agrawal. *Blind Precise Assembly: Proprioceptive Sim-to-Real Peg-in-Hole for Diverse Parts*. Under review for the International Conference on Learning Representations (ICLR), 2027.
- Antonia Bronars*, **Younghyo Park***, and Pulkit Agrawal. *Tune to Learn: How Controller Gains Shape Robot Policy Learning*. Robotics: Science and Systems (RSS), 2026.
- **Younghyo Park**, Jagdeep Bhatia, Lars Ankile, and Pulkit Agrawal. *DART: Dexterous Augmented Reality Teleoperation Platform for Large-Scale Robot Data Collection in Simulation*. IEEE International Conference on Robotics and Automation (ICRA), 2025.
- **Younghyo Park***, Gabriel M. Margolis*, and Pulkit Agrawal. *Position: Automatic Environment Shaping is the Next Frontier in RL*. International Conference on Machine Learning (ICML), 2024. **Oral Presentation, Top 5%.**
- Sunin Kim*, Jaewoon Kwon*, Taeyoon Lee*, **Younghyo Park***, and Julien Perez. *Safety-Aware Unsupervised Skill Discovery*. IEEE International Conference on Robotics and Automation (ICRA), 2023.
- **Younghyo Park***, Seunghun Jeon*, and Taeyoon Lee. *Robot Learning to Paint from Demonstrations*. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2022. **Best Entertainment and Amusement Paper Award.**
- **Younghyo Park**, Joonwoo Ahn, and Jaeheung Park. *Deep Learning Based Parking Slot Detection and Tracking: PSDT-Net*. International Conference on Robot Intelligence Technology and Applications (RiTA 2021), 2022.
- Kyumin Park, **Younghyo Park**, Sangwoong Yoon, and Frank C. Park. *Collision Detection for Robot Manipulators Using Unsupervised Anomaly Detection Algorithms*. IEEE/ASME Transactions on Mechatronics, 2022.

RESEARCH EXPERIENCE

Improbable AI Lab, MIT CSAIL Ph.D. Student

Sep. 2023 – Present

- Advisor: Professor Pulkit Agrawal.
- Study how robot dynamics, controllers, and data-collection interfaces shape learned policies.

NAVER LABS Robotics Researcher

Sep. 2021 – Aug. 2023

- Developed unsupervised skill-discovery methods for manipulation and worked across the robot stack, from hardware and controllers to physical policy execution on systems including AMBIDEX.
- Co-developed ARTO-1, a force-sensitive painting robot exhibited at NAVER 1784; the resulting IROS 2022 paper won the Best Entertainment and Amusement Paper Award, and I am a named inventor on U.S. Patent 12,337,478.

SNU Robotics Laboratory Research Intern

Sep. 2020 – Jan. 2021

- Researched unsupervised collision detection for robot manipulators using anomaly-detection models.

Saige Research Research Intern

Jun. 2020 – Feb. 2021

- Researched out-of-distribution detection for industrial defect imagery.

OPEN SOURCE SOFTWARE

- **VisionProTeleop**: Created and maintain an Apple Vision Pro stack distributed through the App Store and PyPI for physical/simulation teleoperation and egocentric recording; 10,000+ App Store downloads, 800 GitHub stars, 60+ forks, and documented reuse in independent robot-learning systems.
- **aiofranka**: Created and maintain a PyPI-distributed Python stack for Franka robots whose 1 kHz torque-control loop runs in C++, isolated from neural-policy inference; provides impedance and operational-space control, MuJoCo dynamics, smooth trajectories, payload identification, and camera calibration on Linux and Apple Silicon macOS.

- **aioflexiv**: Created a PyPI-distributed counterpart for Flexiv robots, pairing a pybind11 real-time torque shim over flexivrdk with joint-impedance and operational-space control and a MuJoCo backend for simulation and payload-aware dynamics.
- **aprilcube**: Created a 3D-printable fiducial-cube generator and single-image 6-DoF detector; nearly 100 GitHub stars within five months of release.

HONORS & FELLOWSHIPS

Best Paper Awards

- Beyond Teleoperation Workshop, ICRA 2026 — *Tune to Learn: How Controller Gains Shape Robot Policy Learning*.
- Workshop on Manipulation Robustness, ICRA 2026 — *Tune to Learn: How Controller Gains Shape Robot Policy Learning*.
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2022 — *Robot Learning to Paint from Demonstrations*; Best Entertainment and Amusement Paper Award.

Oral Presentations

- CR2 Contact-Rich Robotics Workshop, ICRA 2026 — *Tune to Learn: How Controller Gains Shape Robot Policy Learning*.
- International Conference on Machine Learning (ICML), 2024 — *Position: Automatic Environment Shaping is the Next Frontier in RL*; Oral Presentation, Top 5%.

Fellowships

- Kwanjeong Overseas Fellowship, additional Ph.D. stipend support, 2023–present.
- Merrill Lynch Fellowship, full tuition and stipend, 2023–2024.
- National Science & Technology Scholarship, full undergraduate tuition, 2016–2022.

TEACHING, SERVICE & RESEARCH LEADERSHIP

Workshop Organization

- **Organizer: CoRL 2026 Workshop**. Lead proposer and contact for *Everything Beneath the Policy: How Controllers, Hardware, and Data Infrastructure Shape Robot Learning*; officially accepted.
- **Organizer: Humanoids 2026 Workshop**. *Dexterous Humanoid Manipulation*, bringing together work spanning hardware, control, and autonomy.

Teaching

- **Co-instructor**, MIT 6.S186 *Modern Robot Learning*, IAP 2025. Developed and taught a four-session, project-based course spanning teleoperation, behavior cloning, RL, simulation, and deployment from simulation to hardware; released lectures and materials publicly.

Invited Talks

- IDIM Lab, Seoul National University, Nov. 2025.
- Kimin Lee Lab, KAIST, Aug. 2024.

TECHNICAL SKILLS

Programming: Python, Swift, MATLAB

Libraries and systems: PyTorch, MuJoCo, NVIDIA Isaac Gym/Sim, ROS2

Languages: Korean (native), English